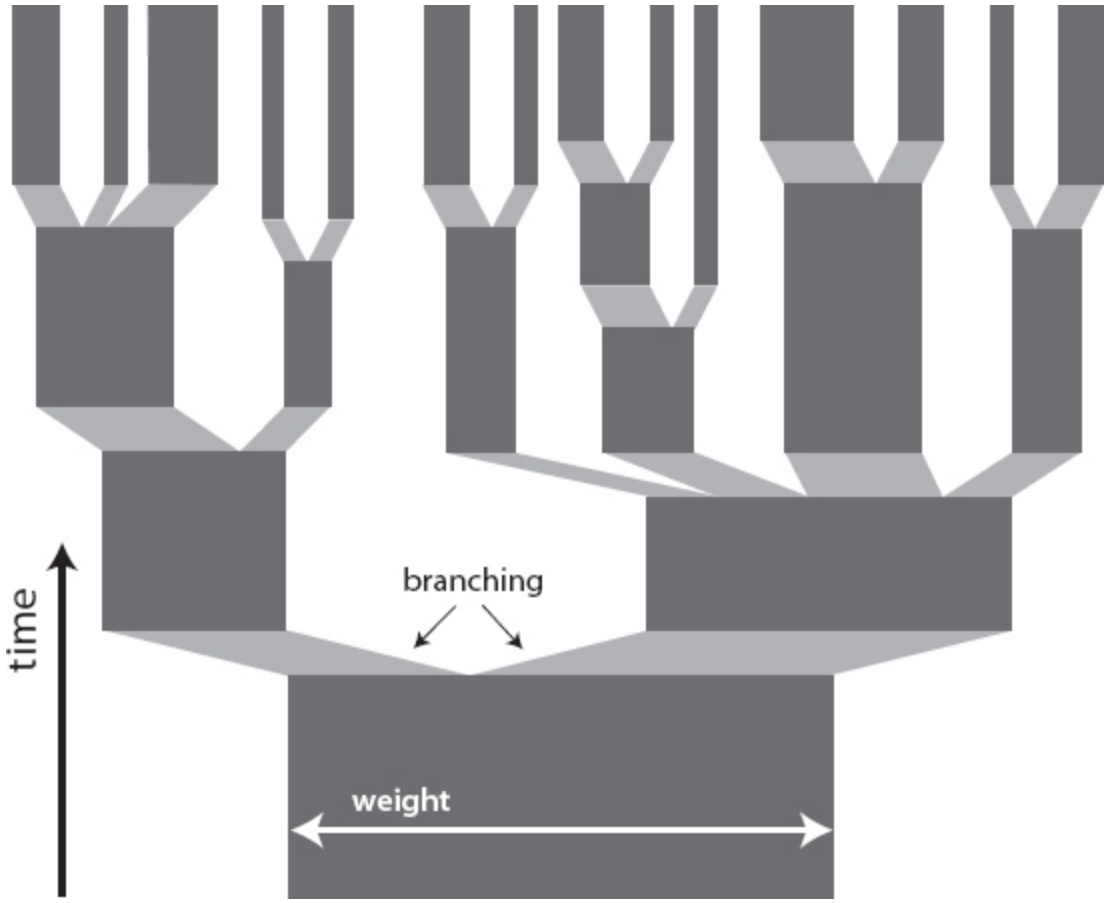


This isn’t the only way to derive the Born rule in the Many-Worlds theory. A strategy that is even more popular in the foundations-of-physics community appeals to decision theory—the rules by which a rational agent makes choices in an uncertain world. This approach was pioneered in 1999 by David Deutsch (one of the physicists who had been impressed by Hugh Everett at the Texas meeting in 1977), and later made more rigorous by David Wallace.

Decision theory posits that rational agents attach different amounts of value, or “utility,” to different things that might happen, and then prefer to maximize the expected amount of utility—the average of all the possible outcomes, weighted by their probabilities. Given two outcomes *A* and *B*, an agent that assigns exactly twice the utility to *B* as to *A* should be indifferent between *A* happening with certainty and *B* happening with 50 percent probability. There are a bunch of reasonable-sounding axioms that any good